

General Relativity

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1 General Relativity

1.1 The Einstein Field Equation

Equivalence Principle: Dynamics Postulate:

Minimal-coupling principle.

Definition 1.1.1 (Spacetime)

Spacetime is an 4-dimensional orientable pseudo-Riemannian manifold (M, g) where g has signature $(3, 1)$.

Definition 1.1.2 (The Einstein Field Equation)

Let (M, g) be spacetime manifold. An Einstein field equation consists of the following data.

- A $(0, 2)$ tensor field T such that $T^{ij} = T^{ji}$ and $\sum_{i=1}^{\dim(M)} \nabla_i T_j^i = 0$ called the stress-energy-momentum tensor.
- A constant K .
- The field equation

$$R_{ij} - \frac{1}{2}Rg_{ij} = \frac{8\pi G}{c^4}T_{ij}$$

where G is the gravitational constant and c is the speed of light.

Definition 1.1.3 (Einstein-Hilbert Action)

1.2 The Schwarzschild Solution